Why Space Has Three Dimensions
A Formal Proof from CRF Axioms 0 and 11

Closing Instability #16:
SO(3) as the Unique Stable Rotation Group
under δ-Operation

Upgrading the hypothesis of CRF_3D_SO3 to a derived result

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Abstract

The document CRF_3D_SO3 established a four-step argument for why space has three dimensions, but identified Step 3 as a hypothesis requiring formal proof: "$n = 3$ is the minimum dimension in which $\mathrm{SO}(n)$ has irreducible representations that are closed under composition and support stable spherical harmonics."

This document closes that gap. The proof proceeds from two axioms only — Axiom 0 and Axiom 11 — without importing any geometric assumption. The key insight is that these two axioms together impose two independent requirements on the rotation group of δ-activity:

1. *Non-abelian* (from Axiom 0: δ never stops, history must accumulate in a path-dependent way, which requires non-commutativity).
2. *Rank 1* (from Axiom 11: the zero-gradient state must be unique, which requires exactly one Casimir operator).

SO(3) is the *unique* rotation group satisfying both conditions. Three spatial dimensions follow.

Contents

1 The Gap Being Closed

CRF_3D_SO3 derived, from Axiom 11 alone:

1. Axiom 11 requires isotropy (derived).
2. Isotropy requires a rotation group SO(n) (derived).
3. $n = 3$ is the minimum stable dimension (hypothesis — the gap).
4. $C \propto 1/r^2$ follows from geometry (derived, conditional on 3).

Step 3 was stated as:

> "The claim that $n \geq 4$ is unstable under δ-operation has not been formally proved. It rests on the observation that SO(n) for $n \geq 4$ has more degrees of freedom — but 'more degrees of freedom' has not been formally connected to 'instability under δ-operation' in CRF's language."

This document provides that formal connection via two lemmas drawing on Axioms 0 and 11 respectively.

2 Definitions

Definition 1 (δ-Stability of a Rotation Group)**.** A rotation group SO(n) is *δ-stable* if and only if:

(S1) **Path-dependence**: The constraint C accumulated through a sequence of R-events depends on the *order* of those events, not only their count.

(S2) **Unique zero-gradient state**: There exists exactly one configuration of SO(n) in which $G = \nabla \Pr(s|C) = 0$ everywhere — the state required by Axiom 11.

Definition 2 (Rank of SO(n))**.** The rank of SO(n) is $\lfloor n/2 \rfloor$ — the number of independent Casimir operators (simultaneously diagonalizable invariants).

3 The Two Lemmas

Lemma 1 (Abelian groups violate S1)**.** SO(n) *satisfies condition S1* if and only if $n \geq 3$.

Proof. $C \equiv \Pr(s|C)$ is self-referential (CRF Session 4). Each R-event updates C multiplicatively: $C_{t+1} = C_t \cdot u_t$ where u_t depends on which state resolves. The accumulated constraint after a sequence of R-events is:

$$C_T = C_0 \cdot u_1 \cdot u_2 \cdots u_T$$

For C_T to encode the *specific history* (which u_t in which order), the multiplication must be *non-commutative*: $u_i u_j \neq u_j u_i$ in general.

In the spatial setting, each u_t corresponds to a rotation in SO(n) — the direction from which the resolving pattern was encountered.

If $\mathrm{SO}(n)$ **is abelian:** $R(a)R(b) = R(a+b) = R(b)R(a)$ for all a, b. The product $u_1 u_2 \cdots u_T$ depends only on the sum $\sum_t u_t$, not on the ordering. C_T cannot distinguish "first x then y" from "first y then x". The specific history is lost.

This violates CRF's core requirement: $t = |R_{\text{events}}|$ and $N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$ both depend on the *specific sequence* of R-events, not just their count.

If $\mathrm{SO}(n)$ **is non-abelian:** The product $u_1 \cdots u_T$ is order-dependent. Different sequences of R-events produce different C_T. History is encoded.

$\mathrm{SO}(n)$ is abelian if and only if $n \leq 2$: $\mathrm{SO}(1) = \{e\}$ (trivial), $\mathrm{SO}(2) \cong U(1)$ (circle group, commutative), $\mathrm{SO}(n)$ for $n \geq 3$ is non-abelian (the Lie algebra $\mathfrak{so}(3) \subset \mathfrak{so}(n)$ has $[J_i, J_j] = \varepsilon_{ijk} J_k \neq 0$).

Therefore S1 holds $\Leftrightarrow n \geq 3$. □

Lemma 2 (High-rank groups violate S2)**.** $\mathrm{SO}(n)$ *satisfies condition S2 if and only if* $n \leq 3$.

Proof. Axiom 11 requires a *unique* zero-gradient state: $G = \nabla \Pr(s|C) = 0$ everywhere has exactly one solution.

In the representation theory of $\mathrm{SO}(n)$, the zero-gradient state is the state in which all Casimir operators take their minimum values (the trivial representation: $J^2 = 0$, $J_1 = J_2 = \cdots = 0$).

The number of *independent* Casimir operators of $\mathrm{SO}(n)$ is $\mathrm{rank}(\mathrm{SO}(n)) = \lfloor n/2 \rfloor$.

For $n \leq 3$: $\lfloor n/2 \rfloor = 1$. There is one Casimir operator J^2. The zero-gradient state is the unique state with $J^2 = 0$: the uniform distribution on S^{n-1}, isotropic in all directions. This is the unique Axiom 11 state. S2 holds.

For $n \geq 4$: $\lfloor n/2 \rfloor \geq 2$. There are at least two independent Casimir operators, say J_1^2 and J_2^2. Each can independently take value zero. The zero-gradient states form a family parameterised by the values of the additional Casimirs:

$$\{(J_1^2 = 0,\ J_2^2 = 0,\ \ldots)\} \quad \text{— all are "zero-gradient"}$$

but they are not all the same distribution. They represent different stable "vacuum" configurations of δ-activity. Axiom 11 requires a *single* zero-gradient state. Multiple such states violate uniqueness. S2 fails.

Therefore S2 holds $\Leftrightarrow n \leq 3$. □

4 The Main Theorem

Theorem 1 (Three Dimensions from δ)**.** *Under Axiom 0 (δ never stops, history accumulates) and Axiom 11 (unique zero-gradient state), the spatial dimension is* $n = 3$.

Proof. By definition, δ-stability requires both S1 and S2.

From Lemma 1: S1 holds $\Leftrightarrow n \geq 3$.

From Lemma 2: S2 holds $\Leftrightarrow n \leq 3$.

Therefore: both S1 and S2 hold $\Leftrightarrow n = 3$.

The case $n = 1$ is excluded separately: $\mathrm{SO}(1) = \{e\}$ is trivial, providing no directions for δ to distinguish — isotropy is vacuous (no structure to be isotropic). Axiom 11 requires a non-trivial stable state, which requires at least one non-trivial direction.

Therefore: $n = 3$ is the unique dimension satisfying both axioms. Three spatial dimensions are derived. □

Corollary 1 ($C \propto 1/r^2$)**.** *In three spatial dimensions, the constraint field C around an isotropic source satisfies $C \propto 1/r^2$.*

Proof. By Gauss's law in 3D (which follows from SO(3) symmetry): the flux of any isotropic field through a sphere of radius r is constant. Surface area $= 4\pi r^2$, so field strength $\propto 1/r^2$. This is a geometric consequence of $n = 3$, which is now derived, not assumed. □

5 The Complete Chain: Axioms to Inverse-Square Law

From δ to $C \propto 1/r^2$ — Fully Derived

$$\text{Axiom 0} \;\Rightarrow\; \text{history path-dependent} \;\Rightarrow\; \mathrm{SO}(n) \text{ non-abelian} \;\Rightarrow\; n \geq 3$$

$$\text{Axiom 11} \;\Rightarrow\; \text{unique zero-gradient state} \;\Rightarrow\; \mathrm{rank}(\mathrm{SO}(n)) = 1 \;\Rightarrow\; n \leq 3$$

$$n \geq 3 \wedge n \leq 3 \;\Rightarrow\; n = 3 \;\Rightarrow\; C \propto \frac{1}{r^2} \;\Rightarrow\; G = \frac{\ln(2) \cdot c^2}{D_0}$$

Every step is forced. No dimension was assumed.

6 What Changes in the CRF Status Table

Result	**Status**	**Note**
Axiom 11 $\Rightarrow$ isotropy	Derived	Prior document
Isotropy $\Rightarrow \mathrm{SO}(n)$	Derived	Prior document
$n = 3$ is unique δ-stable	**Derived**	**This document**
$C \propto 1/r^2$ in 3D	Derived	Corollary 1
Schwarzschild from $C(r)$	Derived	Paper v4
$G = \ln(2) \cdot c^2/D_0$	Derived	Paper v4

Instability #16 (SO(3) minimality) is now closed.

7 Honest Assessment: What This Proof Does and Does Not Do

Scope of the Proof

What is proved:

- Under Axioms 0 and 11, $n = 3$ is the unique dimension.
- The proof uses only representation theory of compact Lie groups (established mathematics) and CRF's axioms.
- No physical space is assumed. The conclusion is that if δ operates under these axioms, the resulting structure must be 3-dimensional.

What is not proved:

- That the Possibility Space P is literally $\mathbb{R}^3$ or S^3. The proof shows that the *symmetry group* is SO(3); the precise topology of P remains open (Instability #1).
- That physical space and δ-space are the same space. The proof derives 3D for the arena of δ-activity; the correspondence with observed physical space is supported by V16 ($d_s = 3.008$) but not formally identified.

CRF Maturity: With Instability #16 closed, the derivation chain from δ to $G = \ln(2) \cdot c^2/D_0$ is complete without any geometric assumption. This advances CRF from Level 6/8 toward Level 7/8. The remaining gate is the P0 experiment.

8 Status Table: Full CRF After This Proof

Result	Status	Source
δ as undefinable primitive	Declared	Axiom 0
$\Pr(s)$ from δ-frequency	Derived	Sessions 1–3
$C \equiv \Pr(s\|C)$	Derived	Session 4
$\theta = 1 - e^{-D_{\mathrm{KL}}/D_0}$	Derived	Session 10
$\alpha = \ln 2$	Derived	Session 8
$n = 3$ from SO(3)	**Derived**	**This document**
$C \propto 1/r^2$	Derived	Corollary 1
Born Rule via Gleason	Theorem	Session 11
Schwarzschild metric	Derived	Paper v4
$G = \ln(2) \cdot c^2/D_0$	Derived	Paper v4
$T_{\mu\nu}$ from C-gradients	Derived	Paper v4
$G(t) \propto \Lambda(t)$	Prediction	Paper v4
Gravity $R^2 = 0.906$	Confirmed	V16
$d_s = 3.008$	Confirmed	V16
Speciation (5 modes)	Confirmed	V18
P0: $\tau_1 < \tau_2$	Testable	SNSPD ready
$C(r)$ exact derivation	Open	#17
Kerr metric	Open	rotating mass

We did not assume three dimensions.
We asked what Axiom 0 and Axiom 11 require together.
Three dimensions answered.
The answer was forced.